# TOOLVERSE

Full-stack Android productivity toolkit

Product requirements and engineering blueprint

**Product promise**

One coherent Android application containing 100+ real utilities for documents, careers, images, data, text, files, security, calculations, QR/barcodes, audio and developer workflows.

**Technology:** Kotlin • Jetpack Compose • Firebase Authentication • Cloud Firestore • Firebase Storage • Cloud Functions • WorkManager • Room • DataStore




Five-screen visual reference supplied in this conversation. Treat this as a design direction; refine generated artifacts into production-grade layouts.

Version 1.0 | 5 November 2026

# 1. Product definition

TOOLVERSE is a native Android full-stack utility platform. It combines focused tools behind one consistent interface, account system, file workflow and processing history. It must not be a webview wrapper or a collection of decorative buttons. Every published tool must perform a real operation, validate inputs, report errors and deliver usable output.

## Product goals

• Deliver 100+ distinct, tested tools across ten categories.

• Make tools discoverable through global search, categories, favourites, recent tools and quick access.

• Support batch workflows, including 50+ images per PDF job, with memory-safe incremental processing.

• Prefer on-device processing for private files; use cloud processing only when needed and with explicit consent.

• Sync user preferences, favourites, drafts and task metadata through Firebase.

• Keep all tools free: no paywalls, subscriptions, premium-only exports or locked actions.

## Non-goals and integrity

• Do not fabricate job listings, employers, salaries, vacancies or application URLs. Career tools create documents from user-provided information.

• Do not claim perfect conversion of every proprietary format; declare supported variants and limitations.

• Do not upload private files by default. Explain cloud upload, retention and deletion before it happens.

**Release rule:** A tool is complete only after input selection, validation, processing, progress, cancellation where feasible, preview, save/share and error states are tested.

# 2. Navigation and application map

| Area | Purpose and contents |
|---|---|
| Home | Greeting, global search, featured utilities, category grid, quick access and recently used tools. |
| Tools | Full catalogue, category filters, alphabetical browse, tool details and favourites. |
| Create/action | Contextual shortcut launcher for common workflows; each task opens its dedicated screen. |
| History | In-progress, completed and failed tasks; open output, retry eligible tasks and remove history. |
| Profile | Account, preferences, appearance, fonts, notifications, privacy, storage, support and logout. |

# 3. Five-screen UI design and implementation




Visual direction: warm cream/off-white surfaces, restrained forest-green primary actions, subtle borders, clear typography, compact category tiles, consistent iconography and modest corner radii. Avoid excessive gradients, 3D illustrations, nested cards, clutter, inconsistent shadows and tiny labels.

## Screen 1 — Welcome and authentication

• Display TOOLVERSE brand, concise value proposition, Continue with Google and Continue with Email.

• No guest mode. No subscription, paywall, premium badge or locked-tool messaging.

• Include Terms and Privacy links, loading state, network error and authentication cancellation state.

• Google authentication uses Firebase Auth provider integration. Email flow supports sign-up, sign-in, verification and password reset.

## Screen 2 — Home dashboard

• Greeting and notification entry; global search across tool names, aliases and categories.

• Featured tool panel with a real tool deep link; popular categories; quick access; recent tools.

• Category counts are generated from the published tool registry, never hard-coded.

• Bottom navigation: Home, Tools, central action, History and Profile.

## Screen 3 — Category listing / PDF tools

• Back navigation, category title, search/filter and concise category description.

• Rows contain consistent icon, tool name, short purpose and navigation affordance.

• Each row opens a dedicated tool workflow. Support scrolling, font scaling and empty states.

## Screen 4 — Tool workflow / Image to PDF

• Use Android Photo Picker and Storage Access Framework; support multiple selection and 50+ images subject to safe device limits.

• Show selected count, thumbnails, reorder, remove and add-more controls.

• Options: A4/Letter/original/custom page size, portrait/landscape, fit/fill, margins, background and output quality.

• Convert incrementally, show real progress, validate generated PDF, preview page count and size, then save/share.

• Never decode all full-resolution images into memory simultaneously.

### Screen 5 — Profile and settings

• Authenticated profile; My Files & History; Favorites; Settings; Appearance; font choice; notifications; Privacy & Security; Help; About; Log Out.

• No Pro badge, upsell, premium tier or subscription controls.

• Provide account deletion and data export entry points.

Responsive implementation: use Compose WindowSizeClass/adaptive layouts. Phones use a readable single-column layout; larger screens may use navigation rail or two-pane layouts. Do not stretch a fixed design until text becomes unreadable.

# 4. Tool catalogue — 121 planned utilities

The inventory below defines the initial scope. Every published entry needs its own workflow, supported-format declaration, validation, output action and automated/manual QA. Shared engines are encouraged; duplicate labels without distinct outcomes are not.

### PDF and document tools — 22

• Image(s) to PDF (50+ images)

• Merge PDF

• Split PDF by page range

• Extract selected PDF pages

• Compress PDF

• Rotate PDF pages

• Reorder PDF pages

• Delete PDF pages

• Duplicate PDF pages

• PDF viewer

• PDF to PNG

• PDF to JPEG

• PDF text extraction

• PDF to DOCX (best effort)

• DOCX to PDF

• TXT to PDF

• HTML to PDF

• Add PDF watermark

• Add page numbers

• Fill PDF form fields

• Sign PDF

• Encrypt/password-protect PDF and decrypt with valid password

### CV, forms and career documents — 16

• CV/resume builder

• Multiple CV templates

• CV profile-photo upload/crop/position

• CV export to PDF

• Resume import and OCR-assisted field extraction

- Cover letter maker
- Job application letter maker
- Offer letter template maker
- Resignation letter maker
- Leave application maker
- Experience certificate template maker
- Recommendation/reference letter maker
- Internship application letter maker
- Professional bio maker
- Fillable form builder
- Form filler with image/signature attachments

## Image and graphics tools — 16

- JPG to PNG
- PNG to JPG/JPEG
- PNG to TIFF
- TIFF to PNG
- TIFF to JPEG
- JPEG to TIFF
- WEBP to PNG
- PNG to WEBP
- HEIC to JPG
- BMP to PNG
- GIF frame extraction
- Resize by pixels
- Resize by percentage
- Crop and rotate
- Image quality/format inspector
- Image metadata viewer/remover

## Data and spreadsheet tools — 12

- CSV maker/editor
- CSV to JSON
- JSON to CSV
- XLSX to CSV
- CSV to XLSX
- JSON formatter/validator
- XML formatter/validator
- YAML formatter/validator
- Duplicate-row remover
- Column/field cleaner
- Delimited-text converter

- Table data to PDF

## Text tools — 12

- TXT maker/editor
- Text to PDF
- Word counter
- Character/line counter
- Case converter
- Whitespace cleaner
- Find and replace
- Text comparison/diff
- Sort lines
- Remove duplicate lines
- Markdown editor/preview
- Text encoding converter

## File and archive tools — 10

- ZIP creator
- ZIP extractor
- 7Z extractor where supported
- Single/batch file renamer
- File type/extension inspector
- File size analyzer
- File organizer by type/date
- Duplicate finder using hashes
- File checksum verifier
- Share/export package creator

## Security tools — 8

- Secure password generator
- SHA-256 calculator
- SHA-1 calculator with legacy warning
- MD5 checksum with integrity-only warning
- File encryption/decryption using authenticated encryption
- Text encryption/decryption
- PDF password protection
- Local cache cleaner and secure-deletion guidance

## Calculators and converters — 10

- Scientific calculator
- Percentage calculator
- Discount calculator
- Tax/GST calculator with configurable rate
- Date difference calculator

• Age calculator

• Time duration calculator

• Length/area/volume converter

• Weight/temperature converter

• Digital storage converter

### QR and barcode tools — 6

• QR generator

• QR scanner

• Barcode generator

• Barcode reader

• Wi-Fi QR generator

• Contact/vCard QR generator

### Audio tools — 5

• Audio format converter for supported codecs

• Audio trimmer

• Audio merger

• Audio metadata viewer/editor

• Audio bitrate/file-size inspector

### Developer tools — 8

• Base64 encoder/decoder

• URL encoder/decoder

• UUID generator

• Unix timestamp converter

• Regex tester

• HTML entity encoder/decoder

• JWT viewer (decode only unless signature verification exists)

• JSON path explorer

# 5. Detailed workflows

## 5.1 Authentication

• Enable Firebase Google and email/password providers only; do not enable anonymous auth.

• Email registration validates input, sends verification and handles duplicate accounts and network failures.

• Email sign-in includes password reset and clear, safe error messages.

• Google sign-in uses Credential Manager / supported Google identity flow and Firebase credential exchange.

• Create minimal user profile in Firestore after authentication; never store passwords in Firestore.

• Support logout, account deletion and data export.

## 5.2 CV/resume builder

• Structured sections: personal/contact details, summary/objective, experience, education, skills, certifications, projects, languages and references.

• Allow add/remove/reorder/hide sections; photo is optional and template-dependent.

• Photo picker supports crop, reposition, replace and remove; preview must match export.

• Offer ATS-friendly templates plus a smaller set of designed templates; support multi-page content and sensible page breaks.

• Export actual PDF using the same template layout as preview; preserve drafts locally and optionally sync metadata.

• Letter makers use user-provided facts and editable templates. Never invent employer, job, salary, qualification or signature details.

### 5.3 Form builder and form filling

• Support text, multiline, date, checkbox, radio, dropdown, image and signature fields.

• Provide field labels, required/optional validation, reorder, preview and save draft.

• Allow images to be attached and positioned in supported output templates.

• Export to PDF; where editing existing PDF form fields is supported, preserve field names and flatten only when requested.

### 5.4 Image conversion

• Use a decoder/encoder capability matrix; expose only format pairs backed by installed and tested libraries.

• Preserve transparency when target format supports alpha. Warn that JPEG has no alpha channel.

• TIFF can be multi-page and use different compression variants; specify exactly which variants are supported.

• Preserve metadata only when requested. Explain metadata removal and privacy implications.

• Do not claim a conversion pair until tested on representative files and Android versions.

### 5.5 PDF and batch processing

• Use incremental page/image processing and bounded memory; do not block main thread.

• Validate source PDFs and report corrupt, encrypted or unsupported documents clearly.

• Use progress updates based on actual work completed, not decorative timers.

• Keep originals unchanged. Write output to a temporary destination first, validate, then finalize export.

• Allow cancellation when safe and clean partial outputs after cancellation/failure.

### 5.6 Job opportunities

If a job-opportunities section is included, connect it to a legitimate job provider or a curated, verified feed with source attribution, timestamps and working application links. Otherwise keep the scope to job-document creation. Never populate a feed with fake demo vacancies.

## 6. Technology and architecture

| Layer | Implementation |
|---|---|
| Language/UI | Kotlin, Jetpack Compose, Material 3, Navigation Compose, adaptive WindowSizeClass. |
| Architecture | Feature-first Clean Architecture; MVVM; StateFlow; coroutines; repositories and use cases. |
| Dependency injection | Hilt, applied consistently. |
| Local storage | Room for drafts/task metadata/cache; DataStore for preferences. |
| Firebase | Firebase Auth, Firestore, Storage, Cloud Functions, App Check, Crashlytics, Analytics and FCM. |
| Background work | WorkManager for persistent/deferrable jobs, constraints, retries and progress; foreground service only where Android policy requires it. |
| File access | Photo Picker, Storage Access Framework and ContentResolver; avoid broad storage permissions without a justified need. |
| Processing | On-device Kotlin/JVM libraries for PDF/image/archive/audio/text; isolated backend worker for heavy cloud jobs. |

### Suggested project modules

`app/` — application setup and navigation
`core/designsystem/` — tokens, components and icons
`core/common/` — result types, validation, dispatchers

`core/data/` — Firebase, Room and repositories
`core/files/` — picker, MIME detection and export
`feature/auth/ home/ tools/ history/ profile/`
`feature/pdf/ cv/ forms/ image/ data/ text/ security/`
`processing/` — engines and format adapters

Each processor should sit behind a domain interface such as ToolProcessor. UI code must not contain conversion logic. Return typed progress, success and failure states.

# 7. Firebase data model

| Path | Fields / purpose | Access |
|---|---|---|
| users/{uid} | displayName, email, photoURL, createdAt, updatedAt, preferences, schemaVersion. | Owner reads/updates allowed fields; protected fields server-controlled. |
| users/{uid}/favorites/{toolId} | toolId, createdAt. | Owner only. |
| users/{uid}/history/{taskId} | toolId, status, inputNames, outputName, sizeBytes, timestamps, errorCode; no file contents. | Owner reads; trusted backend writes protected status. |
| users/{uid}/drafts/{draftId} | draftType, title, structuredData, updatedAt; minimize sensitive fields. | Owner only; validate schema and size. |
| toolCatalog/{toolId} | name, category, description, formats, availability, version, searchAliases. | Public read; service/admin write. |
| processingJobs/{jobId} | uid, toolId, status, progress, storage paths, createdAt, expiresAt. | Owner read; server-only writes. |

Firestore is not a binary file store. Store files locally by default; use Firebase Storage only for explicit cloud workflows or cross-device file sync. Never put file bytes in Firestore documents.

# 8. Privacy and security

• Default-deny Firestore and Storage rules; enforce request.auth.uid ownership for every user resource.

• Enable Firebase App Check and enforce it after validating legitimate traffic.

• Cloud Functions must authenticate and authorize every request, validate MIME type/size and rate-limit expensive work.

• Use authenticated Storage paths such as users/{uid}/jobs/{jobId}/input and output.

• Use HTTPS/TLS. For sensitive cloud files, design encryption and key management before claiming end-to-end encryption.

• Set automatic retention/deletion for temporary files; tell users the retention period and deletion behaviour.

• Never log document text, CV content, passwords, access tokens, private filenames or signed URLs.

• Use Android Keystore-backed encryption for sensitive local drafts where appropriate; keep secrets out of source control.

• PDF encryption must use a vetted library and supported standard; do not store passwords in plaintext.

# 9. Reliability, performance and accessibility

• Use coroutines and bounded dispatchers for CPU-heavy work; never block the main thread.

• Stream or process pages incrementally; enforce memory and disk safeguards.

• Persist job state locally; WorkManager handles eligible retries, constraints and process restarts.

• Provide real progress, cancellation where safe, actionable errors and retry paths.

• Test low-memory devices, Android API range, offline mode, low disk, lost URI permissions and process death.

• Support TalkBack, logical focus, 48dp touch targets, contrast, dynamic font scaling and reduced motion.

• Use lazy lists/grids, stable keys, thumbnail caching and paginated history.

# 10. Design system

| Token | Direction |
|---|---|
| Surface | Warm cream/off-white background; white or near-white cards with subtle borders. |
| Primary | Forest green for primary actions, selected states and progress. |
| Category accents | Muted blue, amber, violet and coral used sparingly to distinguish categories. |
| Typography | Readable sans-serif, strong hierarchy, restrained weights; avoid decorative/condensed faces. |
| Shape | Consistent rounded corners and occasional pill buttons; avoid nested card overload. |
| Elevation/motion | Mostly flat surfaces, subtle shadows only where useful, short purposeful transitions. |
| Themes | Light, dark and system; theme applies consistently to dialogs, sheets, tool screens and system bars. |

Use real vector icons and production typography. Correct any mockup artefacts such as inconsistent status bars, tiny text, artificial illustrations or placeholder labels. Keep the design compact, aligned and calm.

# 11. Error, empty and loading states

- Unsupported format: identify file and list supported alternatives.
- Corrupt/password-protected input: explain the issue; request password only when supported.
- Insufficient storage: explain required space where measurable and allow a different destination.
- Network failure: keep local tools usable; queue/retry eligible cloud tasks.
- Processing failure: preserve source, clean partial output and show a clear reason plus diagnostic reference.
- Empty history/favourites/drafts: concise explanation and direct route to browse tools.
- Expired authentication: request sign-in again without discarding locally saved work.

# 12. QA and acceptance criteria

| Area | Release gate |
|---|---|
| Tool correctness | Each published tool passes positive, negative, boundary and malformed-input tests; output opens in a compatible viewer. |
| Batch files | 50+ image-to-PDF test on low/mid/high memory devices; no OOM crash or main-thread blocking. |
| Authentication | Google/email sign-in, registration, verification, reset, logout, deletion and revoked-session tests. |
| Security | Rules tests for cross-user access, unauthorized job writes, expired links and oversized uploads. |
| UI | Five core screens across compact/normal/large phones, font scaling and light/dark themes. |
| Reliability | Process death, offline, low storage, interrupted upload, retry and cancellation tests. |
| Release | Crash monitoring, privacy disclosures, permissions review, support contact and store listing. |

# 13. Delivery phases

| Phase | Deliverables |
|---|---|
| 0 — Foundation | Gradle/project setup, design tokens, navigation, Firebase environments, CI, security rules. |
| 1 — Core shell | Auth, home, search, categories, tool registry, profile, themes and history framework. |
| 2 — Document essentials | PDF viewer, image-to-PDF, merge/split/compress, TXT-to-PDF, export/share. |
| 3 — Career studio | CV builder, templates, photo, cover/application/offer/resignation letters, form fill and PDF export. |
| 4 — Conversion suite | Image formats, text, data, spreadsheet, archive and metadata tools. |
| 5 — Advanced tools | Security, calculators, QR/barcode, audio and developer utilities. |
| 6 — Hardening | Performance, accessibility, security, device matrix, analytics and release. |

Do not advertise 100+ tools at launch unless at least 100 have passed acceptance tests. It is acceptable to publish completed categories in stages while unfinished tools remain hidden.

# 14. Definition of done

• Clean checkout builds and installs with documented Gradle and Firebase setup.

• Google and email authentication work against real Firebase projects; anonymous/guest login is absent.

• No paywalls, subscription screens, premium labels or locked tool actions exist.

• Every visible tool works with real files and accurately declares supported formats.

• All five reference screens are implemented in Compose with adaptive layouts and shared design tokens.

• File operations preserve originals, produce usable outputs and handle failures safely.

• Cloud functions, Firestore and Storage use production security rules, App Check and retention policies.

• No demo accounts, fake listings, simulated progress, placeholder outputs or fabricated conversion results remain.

# 15. Implementation cautions

Select libraries only after checking current maintenance, Android compatibility, performance and commercial licensing. PDF and office-format conversion are complex; high-fidelity conversions may require a dedicated backend engine. Keep converters behind adapters so the processing engine can change without rewriting UI.

Career templates are editable writing aids, not legal or employment advice. Job listings require a legitimate provider, attribution, timestamps and working application links; otherwise omit the feed.

Document exact file-size limits, codecs, TIFF variants, OCR languages, PDF encryption standards and cloud retention before release. These are tested engineering limits, not marketing promises.

# Appendix — Screen checklist

| Screen | Required elements |
|---|---|
| Welcome | Brand, value proposition, Google/email login, legal links, loading and error states. |
| Home | Greeting, search, featured tool, categories, quick access, recent tools and bottom bar. |
| PDF category | Back, title, search, tool rows, supported-format hints and empty state. |
| Image to PDF | Picker, selected files, reorder/remove/add, options, conversion, progress, preview and export. |
| Profile | Account, history, favourites, appearance, fonts, notifications, privacy, support and logout. |

# Appendix — Product metrics

• Tool completion/failure rate by tool ID and app version.

• Median processing time and memory footprint for local workflows.

• Cloud upload/download success and processing queue latency.

• Crash-free users and ANR rate.

• Search-to-tool-open conversion and most-used categories.

• Privacy-preserving analytics only; never collect document contents or sensitive filenames.

End of TOOLVERSE product and engineering specification.